

# SOCIOL 723: Social Statistics II

Week 12 — Machine Learning: Prediction and Regularization

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# Today

The Bias–Variance Tradeoff

Regularization

Trees and Ensembles

## A Different Goal

Weeks 5–11 asked: *what is the causal effect, and what must be true to recover it?*  
Today we set that aside entirely.

### Prediction versus inference

**Inference:** we care about a parameter  $\beta$ , its interpretation, and its standard error.

**Prediction:** we care only about  $\hat{Y}$  for units we have not seen. The parameters are means, not ends, and may not even be identified.

- These goals conflict. The model that predicts best is usually not the model whose coefficients you would want to interpret — and vice versa.
- Molina and Garip (2019): sociology has plenty of genuine *prediction* problems (who drops out, which neighbourhoods gentrify, which cases to audit), and separate ML tools for describing heterogeneity.
- Next week we put the two back together. Today, keep them apart.

# Bias-Variance

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## Decomposing Prediction Error

Let  $Y = f(X) + \epsilon$  with  $E[\epsilon | X] = 0$  and  $\text{Var}(\epsilon | X) = \sigma^2$ , and let  $\hat{f}$  be estimated from a training sample drawn independently of the new observation. At a new point  $x_0$ , taking expectations over *both* the training sample and the new outcome:

$$E\left[(Y_0 - \hat{f}(x_0))^2\right] = \underbrace{\left(E[\hat{f}(x_0)] - f(x_0)\right)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

- **Bias:** how far the method is wrong on average, across training samples. Rigid models (a straight line) have high bias.
- **Variance:** how much  $\hat{f}$  moves if you redraw the training data. Flexible models (a deep tree) have high variance.
- **Irreducible error:** noise. No method beats it.
- Every tuning parameter in this course is a dial between the first two terms. Penalty size, tree depth, number of neighbours, bandwidth (Week 9!) — all the same dial.

## Why Unbiasedness Stops Being the Goal

Weeks 1–2 prized unbiasedness: Gauss–Markov ranked estimators *within* the unbiased class.

For prediction, that is the wrong criterion

We care about total error. Accepting a little bias to remove a lot of variance lowers MSE. Every method today is deliberately biased.

- OLS with  $k$  regressors has variance growing in  $k$ : with  $p/n$  not small, the variance term swamps everything and OLS predicts badly even though it is unbiased.
- When  $p > n$ , OLS is not even defined —  $\mathbf{X}'\mathbf{X}$  is singular.
- **In-sample fit is not evidence.**  $R^2$  never falls when you add regressors, so training error is a systematically optimistic estimate of test error. The gap is the *optimism*, and it grows with model complexity.

# Estimating Test Error: Cross-Validation

Split the data, fit on part, evaluate on the rest.

## *K*-fold cross-validation

Partition the sample into  $K$  folds. For  $k = 1, \dots, K$ : fit on all folds but  $k$ , predict fold  $k$ , record the error. Average:

$$CV_K = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left( Y_i - \hat{f}^{(-k)}(X_i) \right)^2.$$

- $K = 5$  or  $10$  is standard: leave-one-out ( $K = n$ ) has low bias but high variance and is expensive.
- **Everything data-dependent must happen inside the fold** — variable selection, standardization, imputation. Selecting predictors on the full sample and then cross-validating is a very common and very serious leak.
- With dependence — panels, clusters, time — **split on the independent unit**, not on rows. Same rule as the bootstrap in Week 2.

# Regularization

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# Ridge Regression

Add a penalty on the squared size of the coefficients:

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - X_i' \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad \hat{\beta}^{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{y},$$

where — as always with a penalty —  $\mathbf{X}$  and  $\mathbf{y}$  are centred and the intercept is left out of both the penalty and the displayed matrix.

- Adding  $\lambda \mathbf{I}$  makes the matrix invertible **even when  $p > n$** , which is why ridge works where OLS does not exist.
- Coefficients shrink toward zero but never reach it: ridge does not select.
- **Always standardize the predictors first** — the penalty is not scale-invariant, and the intercept is not penalized.
- Ridge shines under **dense** signals: many small effects, and correlated predictors whose coefficients OLS cannot separate.
- As  $\lambda \rightarrow 0$  we recover OLS; as  $\lambda \rightarrow \infty$  all slopes go to zero and the fit collapses to the sample mean of  $Y$ . The dial is the bias-variance dial.

# Lasso

Penalize the *absolute* size instead:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - X_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

- The  $\ell_1$  penalty sets coefficients exactly to zero, so lasso performs variable selection and estimation simultaneously.
- Geometrically: the  $\ell_1$  constraint region is a diamond with corners on the axes, and the contours of the sum of squares typically first touch it at a corner. The  $\ell_2$  region is a ball with no corners — hence no exact zeros.
- Lasso suits **approximate sparsity**: a few predictors matter, the rest are approximately irrelevant.
- **Elastic net** mixes them,  $\lambda(\alpha \sum |\beta_j| + \frac{1-\alpha}{2} \sum \beta_j^2)$ , which handles groups of correlated predictors better than pure lasso.

## Choosing $\lambda$ , and a Warning About Interpretation

- **Cross-validation** over a grid of  $\lambda$ : pick the minimizer, or the “one standard error” rule — the largest  $\lambda$  within one standard error of the minimum, which buys a simpler model at almost no predictive cost.
- There are also **theoretically derived** penalties that set  $\lambda$  from the model rather than by cross-validation. They matter when the lasso is a *step inside* an inference procedure rather than the final answer.

### Lasso selection is not a variable-importance ranking

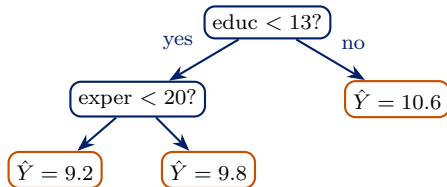
Among correlated predictors, lasso picks one essentially arbitrarily and zeroes the rest. Re-run on a bootstrap sample and you may get a different set. **Do not interpret the selected set as “the variables that matter”**, and do not report post-lasso  $p$ -values from a naive refit — the selection invalidates them. Valid post-selection inference is exactly what Week 13 is for.

# Trees

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## Regression Trees

Recursively split the covariate space to minimize within-node sum of squares; the prediction in a leaf is the mean of the training outcomes there.



- **Strengths:** automatic interactions and nonlinearity; invariant to monotone transformations; handles mixed variable types; readable.
- **Weaknesses:** high variance — a small change in the data can change the first split and hence everything below it. A single deep tree overfits badly; a shallow one underfits.
- Pruning with a complexity penalty helps but does not solve it. The real fix is to average many trees.

## Bagging, Random Forests, Boosting

- **Bagging:** fit trees on bootstrap resamples and average. Averaging  $B$  nearly independent estimates cuts variance by roughly  $1/B$  without adding bias. But bootstrap trees are *correlated* — they all split on the same dominant predictor first — which caps the gain.
- **Random forests:** bag, but at each split consider only a random subset of  $m$  predictors. Decorrelating the trees is what makes the averaging work.  $m$  is the main tuning parameter, alongside node size. Out-of-bag error is a free cross-validation estimate.
- **Boosting:** fit trees *sequentially*, each on the residuals of the current ensemble, adding a shrunk version:  $\hat{f} \leftarrow \hat{f} + \eta \hat{t}$ . Attacks *bias* rather than variance, and with small  $\eta$  and many trees is usually the strongest tabular predictor available.

Bagging and random forests reduce variance; boosting reduces bias. That is the cleanest way to keep them straight.

# What ML Buys, and What It Does Not

## Buys

- Prediction with many predictors, including  $p > n$ .
- Automatic nonlinearity and interactions — no functional form to guess.
- Honest out-of-sample performance estimates.

## Does not buy

- **Causal interpretation.** A variable importance score is not an effect.
- Valid standard errors for the coefficients it selects.
- Extrapolation: trees cannot predict outside the training range at all.
- Freedom from the identification problems of Weeks 5–11.

## The genuine sociological uses

*Prediction as the goal* (screening, triage, forecasting); *measurement* (constructing variables from text or images); *describing heterogeneity*; and **estimating nuisance functions inside a causal estimator** — which is next week.

## Looking Ahead

- Every method today trades bias for variance, tuned by cross-validation.
- That trade is exactly why we cannot drop these estimates into a regression and read off a coefficient: they are *deliberately biased* and converge slowly.
- **Next week:** what has to change before an estimate this biased can be used as an ingredient in a causal estimator — and what it buys us when it can.

### Reading

ISL Chapters 5, 6, and 8; Molina and Garip (2019). Optional: ISL Chapter 10 (deep learning); Kleinberg et al. (2015) on prediction policy problems.



# References

- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning*, 2nd ed. Springer. [Ch. 5, 6, 8]
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